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CodeAlpha - Python Programming Internship
Task 1: Hangman Game

A simple text-based Hangman game where the
player guesses a word
one letter at a time.
"""

import random

# A small list of predefined words
WORDS = ["python", "hangman",
"programming", "internship", "developer"]

MAX_ATTEMPTS = 6


def choose_word():
    """Pick a random word from the word
list."""
    return random.choice(WORDS)


def display_word(word, guessed_letters):
    """Return the word with unguessed
letters shown as underscores."""
    display = ""
    for letter in word:
        if letter in guessed_letters:
            display += letter + " "
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        else:
            display += "_ "
    return display.strip()


def play_hangman():
    word = choose_word()
    guessed_letters = []
    wrong_guesses = 0

    print("=" * 40)
    print("Welcome to Hangman!")
    print(f"You have {MAX_ATTEMPTS}
incorrect guesses allowed.")
    print("=" * 40)

    while wrong_guesses < MAX_ATTEMPTS:
        print("\nWord: " +
display_word(word, guessed_letters))
        print(f"Wrong guesses:
{wrong_guesses}/{MAX_ATTEMPTS}")
        print("Guessed letters:", ",
".join(guessed_letters) if guessed_letters
else "None")

        guess = input("Guess a letter:
").lower().strip()

        # Basic input validation
        if len(guess) != 1 or not
guess.isalpha():
            print("Please enter a single
valid letter.")
            continue
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        if guess in guessed_letters:
            print("You already guessed that
letter. Try again.")
            continue

        guessed_letters.append(guess)

        if guess in word:
            print(f"Good job! '{guess}' is
in the word.")
            # Check if the player has won
            if all(letter in
guessed_letters for letter in word):
                print("\n" + "=" * 40)
                print(f"Congratulations!
You guessed the word: {word}")
                print("=" * 40)
                return
        else:
            wrong_guesses += 1
            print(f"Sorry, '{guess}' is not
in the word.")

    # Player ran out of attempts
    print("\n" + "=" * 40)
    print("Game over! You've run out of
attempts.")
    print(f"The word was: {word}")
    print("=" * 40)


def main():
    while True:
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        play_hangman()
        again = input("\nPlay again? (y/n):
").lower().strip()
        if again != "y":
            print("Thanks for playing.
Goodbye!")
            break


if __name__ == "__main__":
    main()
```